Chen-Chih Wang

Education

Start in Sep 2024 **M.S., Physics, National Tsing Hua University, Hsinchu, Taiwan**

- **Graduation GPA**: 4.25 (on 4.3 scale)
- **Thesis Grade**: A+

Sep 2020 - Jun 2024 **B.S., Physics, National Tsing Hua University, Hsinchu, Taiwan**

- **Graduation GPA**: 4.1 (on 4.3 scale)
- **Ranking**: 5/63

Papers

[1] C.-C. Wang, Y.-P. Huang, and S. Yip, *Exact organization of density matrices and entanglement structure in the Kitaev spin liquid*, Phys. Rev. B **114**, 305105 (2026).

Honors

May 2026 **Honor Member of the Phi Tau Phi Scholastic Honor Society of the Republic of China**, for students who achieved academic excellence upon graduation

Apr 2026 **2026 NTHU College of Science Elite Student Award**, from NTHU for students who show high motivation and exceptional academic performance

Spring 2025 **NTHU Physics Maxwell Scholarship**, for being the top 3 in the Electromagnetic course

Jun 2024 **Honorary Graduate**, from Department of Physics, NTHU, for students with excellent academic performance in undergraduate study

Sep 2023 **Excellent Tutor Award**, from NTHU for outstanding performance as a course tutor (in Counseling Center of NTHU)

Fall 2021, Spring 2022, Fall 2022 **Academic Excellence Award**, from NTHU for being in the top 5% of students in the department

Research Experience

Sep 2024 - Present **Graduate Research**

- **Project**: Density Matrices and Entanglement Properties of the Kitaev Model
- **Advisor**: Prof. Yi-Ping Huang (NTHU)
- Investigated a new way to express the density matrix of the Kitaev ground state, aiming to investigate its deeper entanglement structure

July 2024 - Aug 2024 **Summer Internship, IoP, Academia Sinica**

- **Project**: Kitaev's Projective Solution and the Loop Gas Entangled Spin States
- **Advisor**: Dr. Sungkit Yip (Institute of Physics, Academia Sinica)
- Analyzed the Kitaev ground state in the toric-code limit using parton construction from a coupled-wire perspective
- Awarded the Distinguished Poster Presentation Award

Sep 2022 - Jun 2024 **Undergraduate Research**

- **Project**: Using a quantum entanglement point of view to understand the interaction in a quantum spin liquid
- **Advisor**: Prof. Yi-Ping Huang (NTHU)
- Looking for a tensor network representation of the Kitaev ground state to build the connection between the emergent gauge field and the many-body entanglement pattern
- Awarded the NSTC Undergraduate Student Research Project (Grant: 112-2813-C-007-004-M) [Search]

Oral Presentations

Jan 2025 "Periodic boundary condition of partons in the Kitaev model", 2025 Annual Meeting of the Physical Society of Taiwan (TPS)

Nov 2026 "Exact organization of density matrices and entanglement structure in the Kitaev spin liquid" [slides], Miniworkshop on Quantum Information and CFTs (NTHU)

Poster Presentations

Aug 2026 "Exact Organization of Density Matrices and Entanglement Structure in the Kitaev Spin Liquid" [poster][introduction slides], The 3rd NCTS-iTHEMS Joint Workshop on Theoretical, Mathematical, and Computational Sciences

Jan 2024 "A Green's Function Approach to the $\mathbb{Z}_2$ Gauge Theory in the Kitaev Model", 2024 Annual Meeting of the Physical Society of Taiwan (TPS)

Training

Aug 2025 2025 NCTS Summer School on Frontier Numerical and Analytical Approaches to Low-Dimensional Strongly Correlated Electron Systems

Aug 2024 2024 Strongly correlated physics – Numerical and Analytical approaches

Aug 2023 2023 Tensor Network and Quantum Computing for Strongly Correlated Systems

Teaching Experience

Sep 2024 - Jan 2025 Teaching Assistant, Introduction to Solid State Physics (NTHU)
- **Instructor**: Prof. Tzu-kan Hsiao

Sep 2022 - Present Tutor of Physics, Counseling Center of NTHU

Sep 2021 - Jan 2022 Teaching Assistant, Applied Mathematics, (NTHU)
- **Instructor**: Prof. Martin Spinrath

Programming Skills

Python, MATLAB, LabVIEW, LaTeX

Extracurricular Honors

Mar 2023, Mar 2024, Mar 2025 Distinguished Award, National Student Music Competition (Chinese Silk and Bamboo Ensemble)

Mar 2024, Mar 2025, Mar 2026 Distinguished Award, National Student Music Competition (Chinese Orchestra)

Jan 2025 First Prize, National Student Music Competition, First Round in Hsinchu City (Erhu Solo)